A 3D Medium Operatorial Framework for Environmental Tidal Response

Complete Project Manuscript after Z-line All-Sigma and GR Effective-Source Diagnostic Branch

arXiv-style / Zenodo-ready working paper, V845

Pietro R.
Independent exploratory research project

2026-06-23T12:04:33Z

Not peer reviewed. This manuscript is exploratory and operatorial. It does not claim a physical theory of gravity, a dark matter derivation, a dark energy derivation, an observational detection, or a validated physical field. The V837-V844 branch is a GR effective-source diagnostic branch only. It identifies useful structure and blockers; it does not promote any toy or ansatz to real physics. ENV_RESPONSE remains null-closed with current assets.

Abstract

This V845 manuscript updates the complete project state after the V837-V844 GR effective-source diagnostic branch. The operatorial core and Z-line all-sigma results remain the defensible computational layer. The new branch asks a narrower question: if a compact static spherical toy metric is inserted into standard general relativity, what effective stress-energy source is required, and can that source be interpreted as real matter?

The branch locks the effective-source formulas for $f(r) = 1 - 2Gm(r)/r$, verifies Schwarzschild and de Sitter sanity checks, tests compact mass functions, proves exact interval properties of the S5 phenomenological anisotropic-fluid source, explores a formal magnetic nonlinear-electrodynamics reconstruction, and closes the simple canonical scalar route as a negative control. The result is mixed but useful: S5 is a coherent phenomenological GR diagnostic, but not a physical matter model. NED is formally action-like but blocked for exact compact support by the Maxwell weak-field limit; canonical scalar matter is structurally obstructed. No physical route is reopened.

Contents

1 Executive status after V845 **2**

2 Operatorial result preserved **2**

3 GR effective-source diagnostic route **2**

4 Sanity checks **3**

5 Compact mass-function testbed **3**

6 S5 anisotropic-fluid interval proof **3**

7 NED magnetic reconstruction and blocker **4**

8 Canonical scalar obstruction **4**

9 What the GR effective-source branch establishes **5**

10 GitHub scientific memory guidance **5**

11 Conclusion **6**

A V837-V844 branch table **7**

B Claim boundary table **7**

1 Executive status after V845

The current consolidated status is:

Layer	Status
Baseline operatorial core	**FULL_MATRIX_OPERATORIAL_NO_HARD_FAILS_WITH_WEAK**
V806/V807 baseline	151,128 rows; 150,958 pass; 170 weak; 0 fail
V830 radial-angle null control	151,128 rows; 150,845 pass; 283 weak; 0 fail
Z-line all-sigma	2,099 / 2,099 targets; 40 / 40 V806 runs; 335,840 / 335,840 rows
ENV_RESPONSE	**NULL_CLOSED_WITH_CURRENT_ASSETS**
GR effective-source branch	**DIAGNOSTIC_ONLY**
Physical matter/action	**NOT_SELECTED**
Physical gravity/DM/DE claim	**FORBIDDEN**

2 Operatorial result preserved

The operatorial pipeline remains the defensible core. It maps density-grid inputs to weak-field tidal/electric-Weyl-like proxies, radial environmental summaries, and an auxiliary screened response. The baseline and null-control matrix results remain diagnostic and computational. They are not observations and not a physical theory.

The Z-line all-sigma center-matched matrix remains:

Quantity	Count
Targets covered	2,099 / 2,099
V806 runs completed	40 / 40
Response rows covered	335,840 / 335,840
Coverage status	PASS full coverage

3 GR effective-source diagnostic route

The V837 branch starts from the static spherical one-function metric:

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2, \tag{1}$$

with

$$f(r) = 1 - \frac{2Gm(r)}{r}. \tag{2}$$

In standard GR this corresponds to an effective anisotropic source:

$$\rho_{\text{eff}} = \frac{m'(r)}{4\pi r^2}, \tag{3}$$

$$p_{r,\text{eff}} = -\frac{m'(r)}{4\pi r^2} = -\rho_{\text{eff}}, \tag{4}$$

$$p_{t,\text{eff}} = -\frac{m''(r)}{8\pi r}. \tag{5}$$

This is a reconstruction of the source required by GR. It is not a derivation of a new theory.

4 Sanity checks

V838 verified the formulas against standard checks.

- Schwarzschild vacuum: $m(r) = M$ gives $\rho = p_r = p_t = 0$.
- de Sitter core: $m(r) = \Lambda r^3/(6G)$ gives $\rho = \Lambda/(8\pi G)$ and $p_r = p_t = -\rho$.
- smoothstep compact matching: $m(r) = M(10x^3 - 15x^4 + 6x^5)$ with $x = r/R$ satisfies the intended compact matching at formula level.

5 Compact mass-function testbed

V839 tested three compact families:

Family	Definition	Diagnostic outcome
S3	$3x^2 - 2x^3$	rejected as regular-center negative control
S5	$10x^3 - 15x^4 + 6x^5$	best compact toy candidate, not physical
S7	$35x^4 - 84x^5 + 70x^6 - 20x^7$	smoother but tangential NEC sensitive

The conclusion is that S5 is the best diagnostic toy among those tested, but it is still not a physical matter model.

6 S5 anisotropic-fluid interval proof

For S5:

$$S_5(x) = 10x^3 - 15x^4 + 6x^5, \qquad x = r/R, \tag{6}$$

define

$$A = \frac{15M}{2\pi R^3}. \tag{7}$$

The effective source can be written:

$$\frac{\rho}{A} = (1-x)^2, \tag{8}$$

$$\frac{p_r}{A} = -(1-x)^2, \tag{9}$$

$$\frac{p_t}{A} = -(1-x)(1-2x). \tag{10}$$

V841 proves:

Condition	Verdict
Regular center	PASS
Compact boundary formula	PASS
WEC	PASS
Radial NEC	PASS, saturated
Tangential NEC	PASS
Conservation identity	PASS
SEC	FAIL for $0 \le x < 1/2$
Tangential DEC	FAIL for $2/3 < x < 1$

Thus S5 is a coherent phenomenological anisotropic-fluid diagnostic, but it is not ordinary matter and not fundamental physics.

7 NED magnetic reconstruction and blocker

V842 explored a magnetic nonlinear-electrodynamics route under the convention

$$F = \frac{F_{ab}F^{ab}}{4}, \qquad F(r) = \frac{q^2}{2r^4}. \tag{11}$$

For the magnetic sector:

$$\rho = -L, \qquad p_r = L = -\rho, \qquad p_t = L - 2FL_F. \tag{12}$$

The formal interior reconstruction gives:

$$L(r) = -\rho(r), \tag{13}$$

and

$$L(F) = -A\left(1 - \frac{(q^2/(2F))^{1/4}}{R}\right)^2. \tag{14}$$

The identity $p_t^{\mathrm{NED}} - p_t = 0$ passes formally. However, V843 shows that exact compact support requires

$$L(F) = 0, \qquad L_F(F) = 0 \tag{15}$$

for $0 < F \le F_R$, where $F_R = q^2/(2R^4)$.

This gives $L(F)/F \to 0$ as $F \to 0$, whereas the Maxwell weak-field limit requires $L(F)/F \to -1$. Therefore the exact compact S5 NED route is physically blocked unless the model is changed. A Maxwell-tail alternative breaks exact compact support and creates a different noncompact model.

8 Canonical scalar obstruction

V844 checks the simplest scalar route. For a minimally coupled static canonical scalar in a static non-horizon region $f(r) > 0$, define

$$X = f(r)\phi'(r)^2 \ge 0. \tag{16}$$

The stress pattern is:

$$\rho = \frac{1}{2}X + V, \qquad p_r = \frac{1}{2}X - V, \qquad p_t = -\frac{1}{2}X - V. \tag{17}$$

Therefore:

$$\rho + p_r = X, \qquad \rho + p_t = 0. \tag{18}$$

S5 requires $p_r = -\rho$, hence $\rho + p_r = 0$. In the $f > 0$ region this forces $X = 0$ and therefore $\phi' = 0$. The scalar then becomes isotropic and vacuum-like:

$$p_r = p_t = -\rho. \tag{19}$$

But S5 has

$$\rho + p_t = A\,x(1-x) > 0 \qquad (0 < x < 1). \tag{20}$$

Thus a simple canonical static scalar cannot reproduce the S5 source pattern. A simple phantom sign variant is also obstructed because it still forces $X = 0$. Noncanonical scalar routes are not ruled out, but they remain high-risk and require independent stability gates.

9 What the GR effective-source branch establishes

The V837-V844 branch establishes:

- the toy metric can be interpreted as a GR effective-source diagnostic;
- S5 is the best compact phenomenological source tested so far;
- S5 passes regularity, WEC, NEC, and conservation;
- S5 fails SEC in the inner region and tangential DEC in the outer region;
- NED is formally promising but blocked for exact compact support by the Maxwell limit;
- canonical scalar matter is structurally obstructed.

The branch does not establish:

- a physical matter action;
- a stable field theory;
- local constraints;
- cosmological/lensing consistency;
- a dark matter or dark energy derivation;
- a physical theory of gravity.

10 GitHub scientific memory guidance

A lightweight GitHub repository can be useful as scientific memory, but it should not become a raw-data dump.

Good GitHub content:

- README and claim boundary;
- small scripts from V837-V844;
- compact CSV summaries;
- manifests and SHA256 sidecars;
- LaTeX source and selected small PDFs;
- release notes and reproducibility instructions.

Better as GitHub Releases assets:

- compact ZIP packages V837-V845;
- manuscript PDFs/TEX;
- small recovery packages.

Better for Zenodo or external archival storage:

- large local results;
- raw or heavy derived data bundles;
- 400 MB or multi-GB archives;
- citable scientific data intended for DOI.

Do not publish without review:

- third-party raw data with unclear redistribution rights;
- private paths, usernames, API keys, tokens, or local metadata;
- licensed data not meant for redistribution.

11 Conclusion

The project has made a real methodological advance, not a physical one. The operatorial core and Z-line matrix remain strong diagnostic results. The GR effective-source branch shows that a compact S5 source is mathematically coherent as a phenomenological anisotropic source, but physical interpretation remains blocked. NED fails the compact-support/Maxwell-limit gate for exact compact S5, and canonical scalar matter is structurally obstructed. The correct conclusion remains fail-closed: useful diagnostics, no physical claim.

A V837-V844 branch table

Version	Role	Result	Claim boundary
V837	GR effective stress-energy reconstruction prereg	Locked formulas for f(r)=1-2Gm(r)/r: rho=m'/(4pi r^2), p_r=-rho, p_t=-m"/(8pi r).	GR diagnostic only.
V838	Symbolic sanity checks	Schwarzschild vacuum PASS; de Sitter core PASS; smoothstep compact matching formula PASS.	No matter/action/stability claim.
V839	Compact mass-function testbed	S3 rejected; S5 best compact toy candidate; S7 shows tangential NEC sensitivity.	Toy diagnostic only.
V840	Action/matter model search prereg	Anisotropic fluid ranked first diagnostically; NED ranked highest-risk action-like candidate; canonical scalar ranked obstruction control.	No model selected as physical.
V841	S5 interval energy proof	Regular center PASS, compact boundary formula PASS, WEC PASS, NEC PASS, conservation PASS; SEC fails inner region; tangential DEC fails outer region.	Phenomenological anisotropic fluid diagnostic only.
V842	Magnetic NED formal reconstruction	Formal L(F) reconstruction identity PASS and single-valued interior branch PASS.	Physical NED not selected; blockers remain.
V843	NED Maxwell/threshold blocker	Exact compact support requires L=L_F=0 outside; conflicts with Maxwell weak-field limit; Maxwell tail breaks compact S5.	Exact compact S5 NED physical route blocked.
V844	Canonical scalar obstruction	p_r=-rho forces zero scalar gradient; zero gradient makes isotropic vacuum-like source; S5 anisotropy blocks match.	Canonical scalar route closed as negative control.

B Claim boundary table

Claim	Status	Safe wording
The operatorial core and Z-line all-sigma results remain valid as diagnostics.	Allowed	Operatorial diagnostic results with no physical/observational claim.
The GR effective-source branch converts S5 into a physical matter model.	Forbidden	S5 is a phenomenological GR effective-source diagnostic.
S5 passes regularity, WEC, NEC and conservation.	Allowed	S5 passes selected exact interval diagnostics.
S5 is ordinary matter.	Forbidden	SEC and tangential DEC blockers prevent ordinary matter interpretation.
NED magnetism physically solves the model.	Forbidden	Formal NED reconstruction passes internally but Maxwell/threshold blockers remain.
Canonical scalar is ruled out for this S5 pattern.	Allowed narrowly	Simple static canonical scalar in f>0 region is obstructed; broader noncanonical scalar theories are not ruled out.
ENV_RESPONSE was executed.	Forbidden	ENV_RESPONSE remains null-closed with current assets.
Gravity, DM or DE has been derived.	Forbidden	No physical gravity, dark matter, or dark energy derivation.